

Aaron Hans M. Arboleda

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Professional Summary

A Computer Engineering student with hands-on experience in IT support, hardware assembly, and embedded systems. Proven ability to troubleshoot technical issues and deploy hardware solutions in high-paced environments. Adept at leveraging technical skills to improve system performance and user experience.

Technical Skills

Hardware & Embedded Systems: ESP32, PC assembly, ITX small form factor builds, component diagnostics

Software & Programming: C++, VSCode, Arduino IDE, HTML, JavaScript, Nuxt, Vue, Firebase, MySQL

Soft Skills: Adaptability, Resourceful, Technical Translation, Technical Writing, Cross-Functional Collaboration, Receptive to Feedback, Time Management, Initiative, User-Centric Design

Projects

Smart Sleep Sound Therapy Device with Mobile Application | Research

- Validated the device's effectiveness in improving sleep quality, successfully maintaining a 90% sleep efficiency rate and reducing sleep latency by 66%, through the integration of ESP32 microcontroller, RCWL-0516 radar sensor, and INMP441 microphone.
- Developed a functional sound therapy device, capable of processing ambient noise within milliseconds via MAX98357A Digital Audio Amplifier.

Echolocation Headgear | Digital Signal Processing Project

- Engineered a prototype assistive echolocation wearable device, successfully translating spatial proximity into real-time audio feedback for the visually challenged, by integrating to Arduino Nano with ultrasonic sensors.

Experience

IT Support | SM Megamall Concentrix | July – September 2025 | Mandaluyong City

- Modernized a full production room's infrastructure, ensuring zero downtime during shifts, by executing mass hardware upgrades, software installations, and system cleanups.
- Minimized technical disruptions for call center agents, resolving 50-60 tickets per week, by providing rapid troubleshooting and system diagnostics.

Computer Technician | SDR Computer Trading | July – September 2024 | Quezon City

- Assembled and deployed 25 complete desktop workstations for an educational institution in Palawan, by meticulously selecting and integrating compatible internal hardware components
- Resolved hardware and software malfunctions for an average of 5-10 customers per week, by performing systematic diagnostics and executing component repairs.

Education

Bachelor of Science in Computer Engineering

Polytechnic University of the Philippines | Sta. Mesa Manila, Philippines